

LAB # 6 : Negative Binomial and Geometric Distribution

Introduction

This lab is concerned with two important discrete probability distributions namely negative Binomial and a special case of it known as Geometric distribution. You may find it convenient to print the pdf version of this lab. This lab will take one session.

Negative Binomial Distribution

In the framework of R , the negative binomial distribution describes the number of failed Bernoulli trials that occur before k successes. The probability mass function for such a random variable is

$$f(x) = P(X = x) = \binom{x+k-1}{k-1} p^k q^x$$

where p is the probability of success on an individual trial and $q = 1 - p$ is the probability of failure. The factors of p^k and q^x come from the fact that there must be exactly k successes and x failures. The first $k - 1$ successes must come in the first $x + k - 1$ trials, hence the factor of $\binom{x+k-1}{k-1}$. The last success always comes on the last trial.

Note that there is a different formulation of this distribution when the random variable represents the total number of attempts rather than the number of failures.

The mean of the distribution is kq/p and variance is kq/p^2 .

There are four basic functions in R for calculating the probability in negative binomial distribution.

- `dnbinom(x, k, p)` returns probability mass function $P(X = x)$ with k total failures and probability of success p . As usual x can be vector.

```
dnbinom(3, 4, 0.6)
[1] 0.165888
dnbinom(0:5, 4, 0.6)
[1] 0.12960000 0.20736000 0.20736000 0.16588800 0.11612160 0.07431782
```

- `pnbinom(x, k, p)` returns the cumulative probability $P(X \leq x)$ for negative binomial distribution.

```
pnbinom(3, 4, 0.6)
[1] 0.710208
pnbinom(0:5, 4, 0.6)
[1] 0.1296000 0.3369600 0.5443200 0.7102080 0.8263296 0.9006474
```

- `qnbino(m, p, k, p)` is the inverse cdf of the negative binomial distribution. It returns the smallest value x such that $pnbinom(x, k, p) \geq p$.

```
qnbino(c(0.25, 0.5, 0.75), 4, 0.6)
[1] 1 2 4
pnbinom(0:5, 4, 0.6)
[1] 0.1296000 0.3369600 0.5443200 0.7102080 0.8263296 0.9006474
```

- `rnbinom( $n, k, p$ )` generates n random values from negative binomial distribution with k total successes and probability of success p . Each random number represents failure before fourth success.

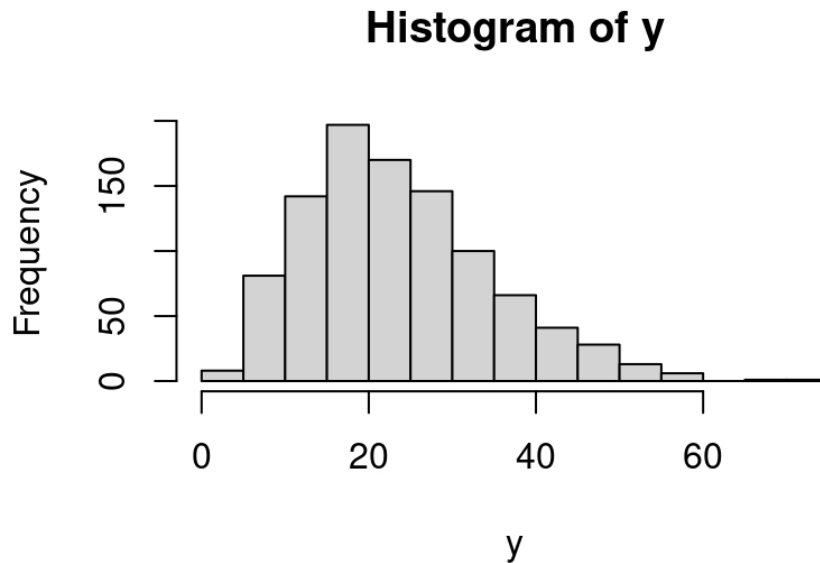
```
rnbinom(10, 4, 0.5)
[1] 7 7 1 2 1 2 2 3 8 4
```

Example 1: A cereal company randomly places 1 of 5 different toys in each box of its cereal. One particular child is trying to collect 6 copies of their favorite toy. This is a negative binomial distribution with $p = 0.2$ and $k = 6$.

- What is the probability that they find the 6th copy of their favorite after finding exactly 25 other toys.
- What is probability that they must open at least 20 disappointing boxes before finding the 6th copy of their favorite toy?
- Simulate the number of failures before the 6th success 1000 times and produce a histogram of the results.

Sol.

- ```
dnbinom(25, 6, 0.2)
[1] 0.03445584
```
- ```
1-pnbinom(19, 6, 0.2)
[1] 0.6166894
```
- ```
y <- rnbinom(1000, 6, 0.2)
hist(y)
```



## Geometric Distribution

The geometric distribution is a special case of the negative binomial distribution with  $k = 1$ . Thus, the distribution models the number of binomial trials until the first success. Let random variable  $X$  to be the number of failures before the first success. The probability mass function is given as

$$f(x) = P(X = x) = q^x p, \quad x = 0, 1, 2, \dots$$

The mean is  $1/p$  and variance is  $q/p^2$ .

There are four basic functions in *R* similar to negative binomial distribution.

- *dgeom*( $x, p$ ) returns  $P(X = x)$  for the distribution where  $x$  is the number of the failures before first success and  $p$  is the probability of success on a given trial. If  $x$  is a vector of integers, then *R* compute the probability for each value in  $x$ .

```
dgeom(5, 0.5)
[1] 0.015625
dgeom(0:4, 0.15)
[1] 0.15000000 0.12750000 0.10837500 0.09211875 0.07830094
```

- *pgeom*( $x, p$ ) returns the cumulative probability  $P(X \leq x)$ . In other words, it provide the arability that at most  $x$  failures occur before the first success.

```
pgeom(5, 0.5)
[1] 0.984375
pgeom(0:4, 0.15)
[1] 0.1500000 0.2775000 0.3858750 0.4779937 0.5562947
```

- *qgeom*( $q, p$ ) returns the smallest integer  $x$  such that  $pgeom(x, p) \geq q$ . This function finds the quantiles in the specified geometric distribution.

```
qgeom(0.7, 0.2)
[1] 5
qgeom(c(0.25, 0.5, 0.75), 0.2)
[1] 1 3 6
pgeom(1:6, 0.2)
[1] 0.3600000 0.4880000 0.5904000 0.6723200 0.7378560 0.7902848
```

- *rgeom*( $n, p$ ) generates  $n$  random numbers form the the geometric distribution.

```
rgeom(10, 0.2)
[1] 6 7 2 5 1 2 2 0 4 12
```

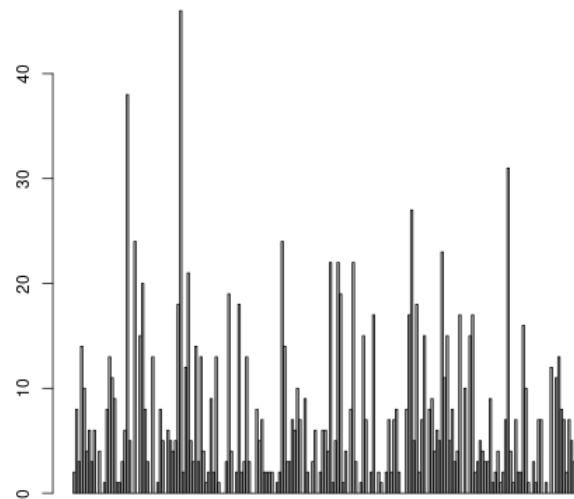
**Example:** According to survey, 13% of Indian don't wear their seatbelt regularly.

- What is the probability that an interviewer calls exactly 8 people before reaching someone who doesn't wears their seatbelt regularly?
- What is the probability that the interviewer calls between 6 and 10 people (inclusive) before reaching someone who doesn't wears their seatbelt regularly?
- A group of 200 interviewer each counts the number of calls before reaching someone who doesn't wears their seatbelt regularly. Simulate the results.

Sol.

```
(a) dgeom(8,0.13)
[1] 0.04266752
(b) pgeom(10,0.13)-pgeom(5,0.13)
[1] 0.2174978
(c) y <- rgeom(200,0.13)
[1] 26 1 2 6 4 7 0 11 1 8 4 5 8 26 12 2 5 1 25 0 1 3 6 3 20
[26] 0 38 2 12 11 14 2 5 1 5 13 7 6 3 13 1 6 8 0 24 6 20 1 5 6
[51] 1 9 3 8 2 2 1 27 5 9 10 12 7 8 4 2 0 3 1 14 5 7 8 5 7
[76] 11 2 12 18 0 12 7 1 10 2 17 0 0 6 6 1 1 8 1 4 14 1 6 7 10
[101] 2 1 9 12 6 7 4 8 6 4 1 6 1 18 9 4 10 5 0 0 24 2 2 19 5
[126] 0 6 2 10 10 8 4 14 8 2 8 4 5 2 2 16 25 7 2 4 5 4 14 8 21
[151] 9 4 1 7 2 4 0 5 19 0 6 12 6 17 20 6 3 14 2 12 3 6 6 2 4
[176] 7 2 4 13 2 4 16 5 0 2 3 9 0 3 3 1 3 14 2 1 1 4 8

summary(y)
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000 2.000 5.000 6.805 9.000 38.000
barplot(y)
```



### Exercices:

- Suppose the probability that any given person will believe a tale about the transgressions of a famous actress is 0.8. What is the probability that
  - the sixth person to hear this tale is the fourth one to believe it?
  - the third person to hear this tale is the first one to believe it?
- In a game, the team that wins four games out of seven is the winner. Suppose that teams *A* and *B* face each other in the championship games and that team *A* has probability 0.55 of winning a game over team *B*.
  - What is the probability that team *A* will win the series in 6 games?

- (b) What is the probability that team  $A$  will win the series?
- (c) If teams  $A$  and  $B$  were facing each other in a regional playoff series, which is decided by winning three out of five games, what is the probability that team  $A$  would win the series?